FIVE EQUIVALENT DESCRIPTIONS OF THE SPECTRAL SUBSETS OF A SPECTRAL SPACE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that the constructible topology of a spectral space is Hausdorff and quasi-compact, and that a closed subset of a quasi-compact space is quasi-compact. Invoked inside the counted proof (source lines 5244-5247). Gives quasi-compactness of an intersection of constructible subsets.
- The statement that a constructible subset meeting the closure of an irreducible subset contains an open of it. Invoked inside the counted proof (source lines 5258-5262). Produces quasi-compact opens containing the subset.

Source reference. Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-constructible-hausdorff-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 4522–4522. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-hausdorff-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-directed-inverse-limit-spectral-spaces: sources/stacks-topology/pinned-original-source.tex, source lines 5119–5119. Navigation pointer only: the counted text refers to the Stacks item lemma-directed-inverse-limit-spectral-spaces, which is an existing labeled result of this same chapter and is

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Source reference. Original lemma lemma-intersection-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1941–1941. Navigation pointer only: the counted text refers to the Stacks item lemma-intersection-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let W be a subset of a spectral space X. The following are equivalent:*

1. *W is an intersection of constructible sets and closed under generalizations,*
2. *W is quasi-compact and closed under generalizations,*
3. *there exists a quasi-compact subset $E \subset X$ such that W is the set of points specializing to E,*
4. *W is an intersection of quasi-compact open subsets,*
5. *there exists a nonempty set I and quasi-compact opens $U_i \subset X$, $i \in I$ such that $W = \bigcap U_i$ and for all $i, j \in I$ there exists a $k \in I$ with $U_k \subset U_i \cap U_j$.*

In this case we have (a) W is a spectral space, (b) $W = \lim U_i$ as topological spaces, and (c) for any open U containing W there exists an i with $U_i \subset U$.

Proof. Let $W \subset X$ satisfy (1). Then W is closed in the constructible topology, hence quasi-compact in the constructible topology (by Lemmas lemma-constructible-hausdorff-quasi-compact and lemma-closed-in-quasi-compact), hence quasi-compact in the topology of X (because opens in X are open in the constructible topology). Thus (2) holds.

It is clear that (2) implies (3) by taking $E = W$.

Let X be a spectral space and let $E \subset W$ be as in (3). Since every point of W specializes to a point of E we see that an open of W which contains E is equal to W. Hence since E is quasi-compact, so is W. If $x \in X$, $x \notin W$, then $Z = \overline{\{x\}}$ is disjoint from W. Since W is quasi-compact we can find a quasi-compact open U with $W \subset U$ and $U \cap Z = \emptyset$. We conclude that (4) holds.

If $W = \bigcap_{j \in J} U_j$ then setting I equal to the set of finite subsets of J and $U_i = U_{j_1} \cap \ldots \cap U_{j_r}$ for $i = \{j_1, \ldots, j_r\}$ shows that (4) implies (5). It is immediate that (5) implies (1).

Let I and U_i be as in (5). Since $W = \bigcap U_i$ we have $W = \lim U_i$ by the universal property of limits. Then W is a spectral space by Lemma lemma-directed-inverse-limit-spectral-spaces. Let $U \subset X$ be an open neighbourhood of W. Then $E_i = U_i \cap (X \setminus U)$ is a family of constructible subsets of the spectral space $Z = X \setminus U$ with empty intersection. Using that the spectral topology on Z is quasi-compact (Lemma lemma-constructible-hausdorff-quasi-compact) we conclude from Lemma lemma-intersection-closed-in-quasi-compact that $E_i = \emptyset$ for some i. □